

Mohit Katre

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TECHNICAL SKILLS

- **Programming Languages:** Python, Java, TypeScript, SQL, C
- **Backend & Web:** FastAPI, RESTful APIs, React, PostgreSQL, Redis, JWT Authentication
- **Cloud & DevOps:** Amazon Web Services (AWS: EC2, S3, IAM), Terraform (Infrastructure as Code), Docker, Docker Compose, Nginx, GitHub Actions, CI/CD
- **Data & Analytics:** Power BI, DAX, Power Query, Apache Kafka
- **Embedded & Systems:** Raspberry Pi, Linux, Bash/Shell Scripting, Git, GitHub
- **Coursework:** Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems, Computer Networks

EDUCATION

Shri Ramdeobaba College of Engineering and Management (RCOEM)

Nagpur, India

Bachelor of Technology (B.Tech.), Electronics & Communication Engineering

Aug 2023 - Jun 2027

CGPA: 7.49/10

EXPERIENCE

Sarvaksh Communications Technologies Pvt. Ltd.

Nagpur, India

Embedded Software Engineering Intern

Jan 2026 - Apr 2026

- Developed a Li-Fi optical wireless communication prototype on Raspberry Pi 5 that streams real-time video over visible light, using Python, Linux and Bash to achieve end-to-end latency under **50 ms**.
- Implemented and tuned an On-Off Keying (OOK) modulation protocol, reaching **95% reliable data transmission** over 1-2 meters through iterative testing, debugging and signal-quality analysis.
- Reduced signal and synchronization failures by **40%** by diagnosing hardware-software timing and firmware issues through root-cause analysis alongside a cross-functional engineering team.

PROJECTS

OptiStock | Multi-Tenant Inventory Intelligence Platform

Python, FastAPI, PostgreSQL, Redis Streams, React, TypeScript, Docker, AWS

- Architected an event-driven inventory platform with **82 REST endpoints** across **24 PostgreSQL tables**, using the Transactional Outbox Pattern and Redis Streams for reliable, idempotent event delivery.
- Built a demand-planning engine from scratch that analyzes **3 years** of transaction history nightly to drive ABC classification, demand forecasting and reorder-point recommendations.
- Shipped a Gemini-powered AI assistant with SSE streaming, tenant-isolated tools and JWT authentication, validated by **465 automated tests** and deployed on AWS EC2 using Docker Compose and Nginx.

Castle | Cloud Deployment Platform

AWS, Terraform, TypeScript, React

- Built a Vercel-style deployment platform where every Git push triggers an isolated container build served through a CDN, provisioning **about 20 AWS services** entirely as code with Terraform.
- Designed immutable builds with the live URL acting as a pointer, enabling rollback to any previous version in **under 1 second** without rebuilding.
- Secured untrusted builds with per-build, least-privilege credentials scoped to a single S3 prefix, and operated the full platform on AWS for **about \$2 per month**.

CERTIFICATIONS & PUBLICATIONS

- **AWS Certified Cloud Practitioner (CLF-C02):** Cloud architecture, EC2, S3, IAM, security.
- **AWS Certified AI Practitioner (AIF-C01):** Generative AI, AWS AI/ML services, responsible AI.
- **IEEE Research Publication:** Random Forest-based urban land-cover classification using Landsat 8/9 satellite imagery with uncertainty analysis.

LEADERSHIP & ACHIEVEMENTS

- Solved **300+** data structures and algorithms problems across LeetCode, HackerRank and CodeChef.
- Chaired **3** Model United Nations (MUN) UNGA committees and spoke at **20+** public forums, moderating debates and building consensus among delegates.
- Led a **100+** member Spiritual Club as Management & Content Coordinator, organizing **5+** events and creating content that kept members informed of club activities.